Electricity

Charge & circuits

1 Current, pd & resistance

Q = It; V = IR. Current = charge per second. Pd = energy per coulomb.

A complete circuit needs a source of pd. Charge is not used up. Ammeter in series; voltmeter in parallel.

2 Series vs parallel

Series: same I, shared V, **$R_{total} = R_1 + R_2$.**

Parallel: same V, branch currents add. More paths reduce total resistance below the smallest branch R.

Measure behaviour

3 Components

Ohmic: constant-temperaturestraight I–V line. Lamp heats → R rises. Diode conducts one way.

Thermistor R falls when hotter. LDR R falls when brighter; both can be sensors.

4 Circuit practicals

Wire: vary length, measure V/I; fix material, thickness and temperature. Switch off between readings.

I–V: vary supply, record paired values, reverse polarity; use protective resistor with diode.

Transfer & protect

5 Power, mains & Grid

$P = VI = I^2R$; $E = Pt = QV$. UK mains: 230 V, 50 Hz ac.

Live brown; neutral blue; earth green/yellow. Grid: higher V → lower I at same power → less I^2R heating.

6 Static & fields

Positive sphere: field points out

Rubbing transfers electrons: gain → negative; loss → positive. Like repel; unlike attract.

Field acts on charges without contact. Strong fields can ionise air → spark.